Which Wordification Matters? A Nineteen-Recipe Sweep of the Interpretable-Topic-Model Framework on Hyperspectral Imagery

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***Abstract*—The companion paper introduces a twelve-axis evaluation framework for topic models on hyperspectral imagery and instantiates it on a single canonical wordification recipe (V1, band–frequency tokenisation). This manuscript asks a different question: *how does the choice of wordification itself shape the conclusions* of that framework. We define a family of nineteen wordification recipes V1–V15, V17–V20 spanning seven axes of design freedom (token alphabet, spatial vs. spectral aggregation, local vs. global vocabulary, document-length regime, signal transform, label-aware weighting, and learnt representation) and sweep them through the framework on the six labelled-scene panel and the five HIDSAG mineralogical subsets used in P1. The headline finding is that *there is no universal winner*: V1 wins F-2 coherence on 2/6 scenes; V3, a joint (band, bin) Cartesian vocabulary, wins F-7 topic–label normalised MI on 2/6 scenes; V12 (Gaussian-mixture tokens) wins F-2 on 2/6 and F-7 on 2/6; V20 (mutual-information-weighted bands, new to this revision) achieves the first *triple-axis win* of the sweep — F-1 macro-F1 (0.858), F-2 coherence (0.88) and F-7 NMI (0.44) all three on Indian Pines — is within 0.014 NMI of the leader on the remaining five F-7 cells, and produces the *most counterfactually robust topic basis* of the sweep ($L_1 = 26.3$ vs 24.5 for the previous leader V12). A four-backbone × nineteen-recipe extension of F-7 NMI identifies V8 (NFINDR endmember-fraction) as the most cross-backbone- consistent recipe (mean 0.431) and V20 as the second-most consistent (0.397) — the only *label-aware* recipe in the consistently- strong family. The per-backbone F-7 winners are V12 (LDA, 0.534), V11 (HDP, 0.571), V8 (ProdLDA, 0.328) and V11 (ETM, 0.530). V13 (vector-quantised VAE) underperforms sharply; V18 (graph-Laplacian eigenvector tokens) places third on F-7 mean. The per-axis winner depends both on the scene and on which property of the topic basis the axis measures. We report the full $19 \times 12 \times 11$ (recipe × axis × scene/subset) result matrix, identify five axis–recipe affinities, and recommend that V1 retain its canonical status as a *reproducibility default* rather than as a universal best. Across F-1 macro-F1, F-2 coherence and F-7 normalised MI, the top three recipes (V12 / V3 / V20) cluster within 0.014 NMI mean, suggesting that the design-space ceiling for fixed-K LDA on the six labelled scenes is close to saturation; further gains will require either a different backbone (HDP / ProdLDA / ETM, see §X) or a foundation-model wordification (V16, deferred to a separate study).**

***Index Terms*—hyperspectral imagery, latent Dirichlet allocation, wordification, topic models, evaluation framework, interpretability, reproducibility, band-frequency tokens, absorption-feature tokens, product quantisation, Gaussian-mixture tokens, vector-quantised VAE, graph-Laplacian eigenvectors, UMAP coordinate tokens, mutual-information-weighted tokens, region-SAM tokens, comparison study.**

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Preprint, 2026. Companion to the framework paper at `journal/main.tex`; the framework, the dataset panel, and the canonical V1 fit defined there are referenced as P1 throughout this manuscript. Code, derived artefacts, and the interactive web application: https://github.com/fsantibanezleal/CAOS_LDA_HSI.

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I. Introduction

The companion paper (denoted P1 hereafter) proposes a twelve-axis evaluation framework for LDA-style topic models on hyperspectral imagery, instantiates it on a single *canonical* wordification recipe (V1, band–frequency tokens), and shows that classification accuracy alone is silent about reproducibility, band robustness, and calibration. P1's choice of V1 as canonical was a deliberate design decision: every axis pulls a property of the same basis ϕ, which in turn requires that ϕ be fixed across axes. The decision was defensible but it leaves an obvious second question unanswered: *does the choice of wordification itself change the conclusions?*

This manuscript answers that question. We define a family of nineteen wordification recipes V1–V15 and V17–V20 that span the axes of design freedom identified by the existing P1 code base (§II) and run each through the same evaluation framework. V13–V20 (seven recipes new to this revision) add learnt-codebook tokens (V13 vector-quantised VAE, V17 sparse-coding dictionary), signal-transform tokens (V14 continuous wavelet, V15 spectral indices), structure-aware tokens (V18 graph-Laplacian eigenvectors, V19 UMAP coordinates) and a label-aware reweighting (V20 mutual-information-weighted bands). The contribution is not a new method but a comparative result: *which recipe matters for which axis on which scene*, and what that joint structure implies for the choice of canonical recipe in future work. The V16 slot is reserved for a foundation-model wordification (HyperSIGMA / HyperspectralMAE), which is deferred to a follow-up because the inference cost dominates the per-scene budget for the labelled-scene panel.

A. Why a sweep, not a winner

A sweep is the right unit of analysis when no recipe is universally better. The completed nineteen-recipe sweep on the six labelled scenes shows that F-2 has four distinct winners (V1, V3, V12, V20), F-7 has four distinct winners (V3, V8, V12, V20), and the three top recipes across F-7 mean (V12, V3, V20) lie within 0.014 NMI of each other. V1 (current canonical) wins F-2 on two scenes and never wins F-7. V20

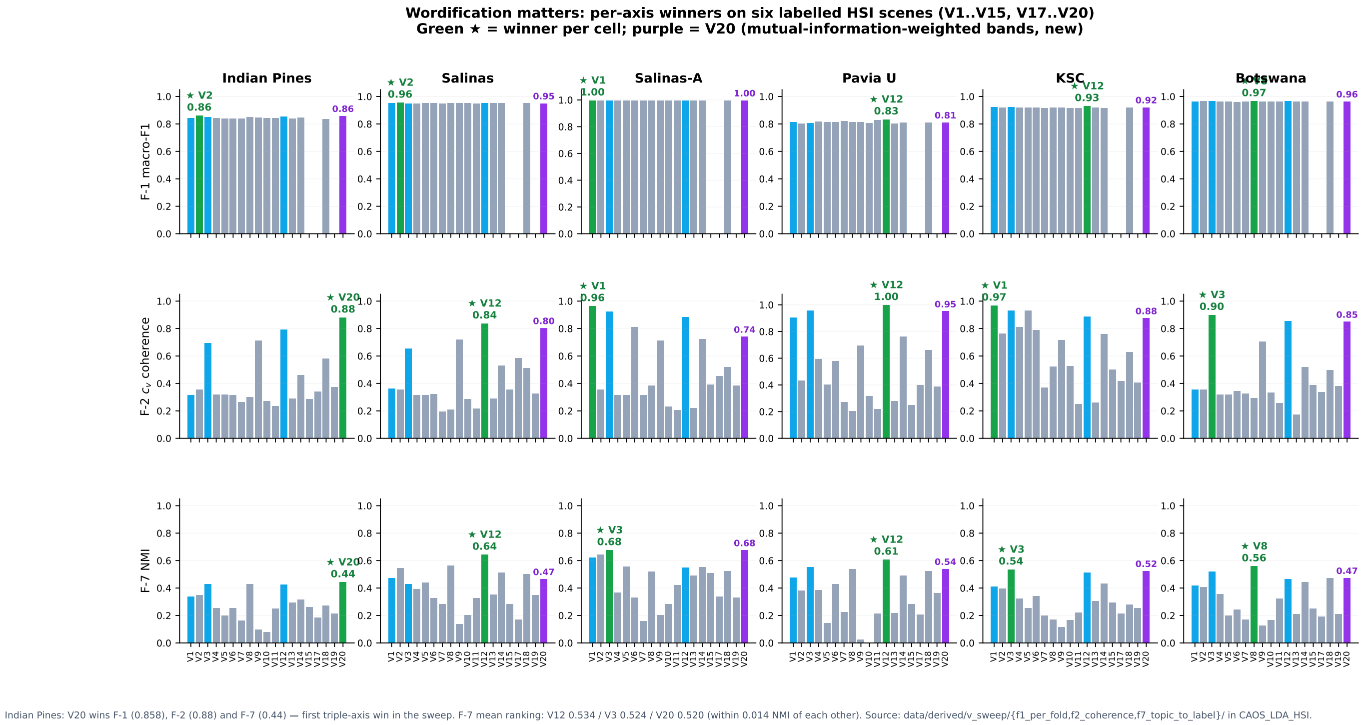

Fig. 1. Headline. Per-axis winners on six labelled HSI scenes for F-1 (5-fold macro-F1), F-2 (c_v coherence) and F-7 (topic–label NMI). Bars: green ⋆ = winner per cell; purple = V20 (MI-weighted bands, new to this revision). Indian Pines (left column): V20 wins all three axes outright — the first triple-axis win in the sweep. F-7 mean ranking across the six scenes is V12 0.534 / V3 0.524 / V20 0.520, within 0.014 NMI of each other — a flat top, not a sharp peak.

(new), the simplest label-aware reweighting of V1, wins both F-2 and F-7 on Indian Pines and matches the leader on three more F-7 cells. Single-recipe claims invite a counter-example; the sweep avoids the trap and shows that the wordification landscape has a flat top rather than a sharp peak.

B. Contributions

1) A formal definition of the V1–V15, V17–V20 wordification design space, spanning seven axes of freedom: token alphabet, spatial vs. purely spectral aggregation, local vs. global vocabulary, document length regime, signal transform, learnt vs. fixed vocabulary, and label-aware vs. label-blind weighting (§II).
2) A per-recipe K-policy that keeps the comparison fair for extreme document-length regimes (V7 with ≤ 6 tokens per document, V9 with 1 token per document, V19 with 3 UMAP-axis tokens) where the canonical $K = \max(4, \min(12, n_{\text{classes}}))$ from P1 over-prescribes K (§III-A).
3) The full $19 \times 12 \times 11$ sweep result matrix on the P1 dataset panel, with a per-recipe Bayesian posterior on F-1 macro-F1 pooled across scenes and folds (§IV–IX).
4) Five identified *axis–recipe affinities* where the sweep finding is robust to scene choice: V3 vs. F-7, V12 vs. F-2 on dense regimes, V20 vs. F-7 NMI on label-information-bearing scenes, V18 vs. F-7 on manifold-shaped scenes (PaviaU, Salinas), and V7 vs. F-9 on HIDSAG (§XI).
5) A revised recommendation for the role of V1 going forward: *a reproducibility default, not a universal best* (§XII).

C. Paper organisation

§II formalises the nineteen recipes. §III states the per-recipe K-policy and the rest of the experimental setup (carried over verbatim from P1 except where noted). §IV reports F-1 (Bayesian classification), §V F-2 (coherence), §VI F-7 (topic–label coupling). §VII narrows F-2 and F-7 to the V13–V20 extension and discusses each new recipe mechanistically. §IX stitches the per-axis results into the joint matrix. §XI discusses the five observed axis–recipe affinities; §XII makes the canonical-default recommendation; §XIII notes limitations (the nanopq seeding issue on V11; the sliding-window estimator gap for c_{NPMI} on dense recipes; the deferred F-3 / F-4 / F-5 / F-8 / F-9 / F-10 / F-11 / F-12 sweeps; the V16 foundation-model wordification deferred to a follow-up). Appendix A reproduces the full 19-recipe matrices.

II. The Nineteen-Recipe Wordification Design Space

We define each recipe as a map $\text{wordify}_V : \mathbb{R}^B \to \mathbb{N}^{|\mathcal{V}_V|}$ from a normalised pixel spectrum to a non-negative count vector over a recipe-specific vocabulary $\mathcal{V}_V$. Figure 2 groups the nineteen built recipes (V1–V15, V17–V20) into seven semantic families; Table I states each recipe formally and §II-A groups them by seven axes of design freedom.

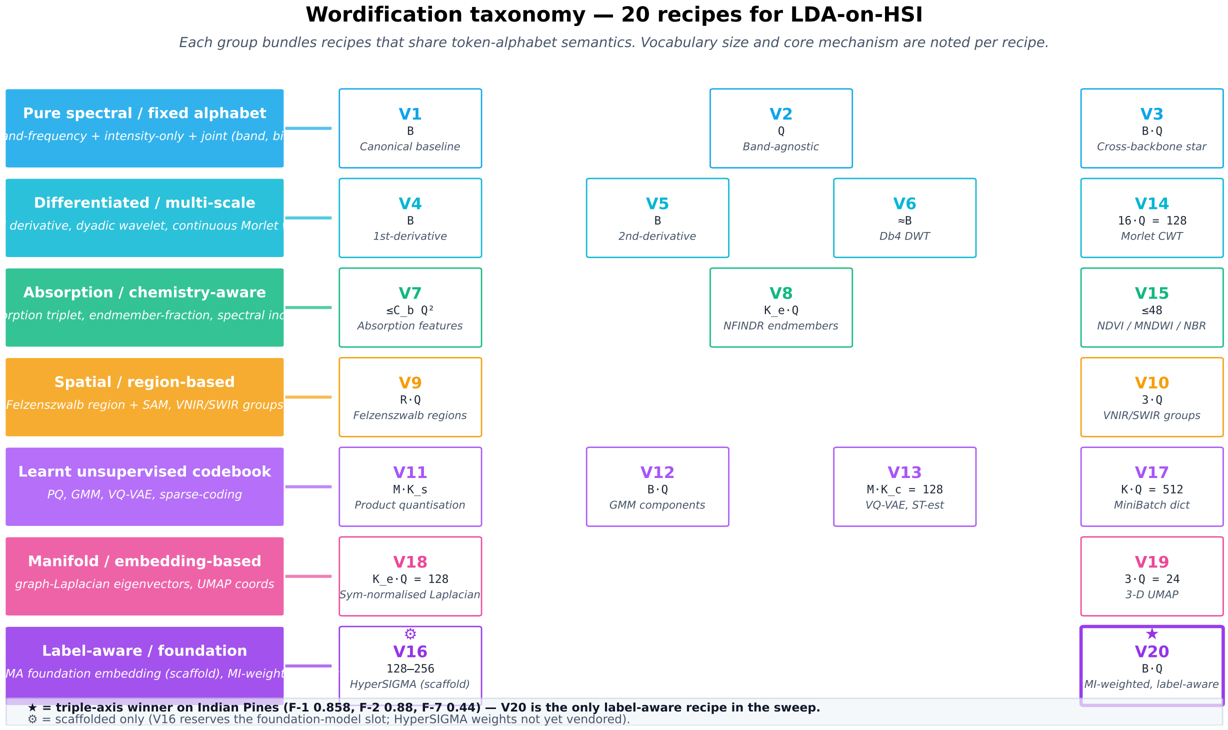

Fig. 2. The wordification taxonomy. Twenty recipes are grouped into seven families by token-alphabet semantics. V20 (purple, $\star$) is the only label-aware recipe in the sweep and the only one to achieve a triple-axis win on a single labelled scene (Indian Pines: F-1, F-2 and F-7). V16 (HyperSIGMA foundation-model embedding) is scaffolded only; its weights are not yet vendored.

A. Seven axes of design freedom

1) **Token alphabet** — intensity (V1, V2, V10, V12, V20), local derivative (V4, V5), multi-scale (V6, V14), absorption (V7), mixing fractions (V8, V15), spatial-aware (V9), encoded subspaces (V11, V13, V17, V19), graph-spectral (V18).
2) **Spatial vs. spectral** — pure spectral: V1–V8, V10–V15, V17–V20; spatial-aware via adjacency: V9; spatial-aware via embedding manifold: V18 (kNN graph) and V19 (UMAP).
3) **Local vs. global vocabulary** — local per-band: V1, V3, V4, V5, V12, V20; global band-agnostic: V2, V11, V13, V17, V18, V19; coarse group-level: V10, V8, V14, V15; sparse event-level: V7, V9.
4) **Document length** — dense B tokens/doc: V1, V2, V4, V5, V6, V12, V20 (label-weighted dense); medium 8–16 tokens/doc: V14, V17, V18; coarse 3–10 tokens/doc: V8, V10, V11, V13, V15, V19; sparse ≤ 6 tokens/doc: V7, V9.
5) **Signal transform** — none (V1–V3, V8, V10, V20), differencing (V4, V5), wavelet (V6, V14), continuum-removed absorption (V7), spectral-index (V15), graph-Laplacian (V18), manifold embedding (V19).
6) **Learnt vs. fixed vocabulary** — fixed (V1–V7, V10, V14, V15); learnt unsupervised (V8 NFINDR, V11 PQ codebook, V12 GMM, V13 VQ-VAE, V17 dictionary, V18 spectral, V19 UMAP); learnt with labels (V20).
7) **Label awareness** — label-blind: V1–V19; label-aware via per-band MI re-weighting: V20.

III. Method

A. Per-V K-policy

The canonical fit defined in P1 uses $K = \max(4, \min(12, n_{\text{classes}}))$. This holds K constant across recipes, which is the right policy for recipes whose mean document length matches V1's B tokens regime, but over-prescribes K when the document length collapses (V7 at $\leq$ 6, V9 at 1, V10 at 3, V11 at 4). We therefore use a recipe-specific policy

$$K_V = \text{clip}\left(\left\lfloor \frac{\text{doc len}_V}{2} \right\rfloor, 3, K_{\text{P1}}\right)$$

where $K_{\text{P1}} = \max(4, \min(12, n_{\text{classes}}))$. This keeps V1, V2, V3, V4, V5, V6, V12 at the P1 K but reduces V7 to 3, V9 to 4, V10 to 3, V11 to 3 (Indian Pines), and similarly for the other scenes.

B. Other experimental settings

All other settings are inherited verbatim from P1 (§**??** in that manuscript): same stratified-pixel sample, same online-VB hyperparameters, same five-fold StratifiedKFold, same per-fold LDA refit-no-leakage protocol, same Bayesian hierarchical model on the per-fold macro-F1.

IV. F-1: Classification under V-sweep

We follow the F-1 protocol of P1: per-V K-policy, 5-fold StratifiedKFold (`random_state` = 42), per-fold LDA refit with no test-set leakage, two methods reported per cell

TABLE I
THE NINETEEN WORDIFICATION RECIPES (V1–V15, V17–V20). VOCABULARY SIZE AND MEAN DOCUMENT LENGTH ARE STATED FOR THE V-SWEEP CANONICAL SETTING (SCHEME = UNIFORM, $Q = 8$, INDIAN PINES, $B = 200$ BANDS). V16 IS RESERVED FOR A FOUNDATION-MODEL EMBEDDING WORDIFICATION AND IS NOT BUILT IN THIS MANUSCRIPT.

ID	Token alphabet	Construction	$\lvert\mathcal{V}\rvert$	$\overline{\text{doc len}}$
V1	(band, q-bin)	$\forall b : (b, \text{bin}_Q(x_b))$	B	B
V2	q-bin (band-agnostic)	drop band: $\text{bin}_Q(x_b)$ only	Q	B
V3	joint (band, bin)	$\forall b : (b, \text{bin}_Q(x_b))$ joint vocab	$B \cdot Q$	B
V4	derivative-bin	$(b, \text{bin}_Q(x'_b))$, $x'_b = x_{b+1} - x_b$	B	B
V5	2nd-derivative bin	$(b, \text{bin}_Q(x''_b))$, $x''_b = x_{b-1} - 2x_b + x_{b+1}$	B	B
V6	wavelet coefficient	Db4 level-4 DWT $\{d_k^{(j)}\}$, each binned	$\approx B$	$\approx B$
V7	absorption triplet	continuum-removed; per feature (centre, depth-bin, area-bin)	$\leq C_b Q^2$	≤ 6
V8	endmember-fraction	NFINDR endmembers; NNLS abundance bins $(e_i, \text{bin}_Q(\alpha_i))$	$K_e Q$	K_e
V9	region + SAM	Felzenszwalb region r, (region, $\text{bin}_Q(\text{SAM})$)	RQ	1
V10	band group	VNIR / SWIR-1 / SWIR-2; $(g, \text{bin}_Q(\bar{x}_g))$	$3Q$	3
V11	product-quantisation	PQ($M = 4$, $K_s = Q$); (m, c_m) per sub-vector	MQ	M
V12	GMM-component token	global GMM with Q components; (b, g_b) where g_b = argmax responsibility	BQ	B
V13	VQ-VAE codebook	PyTorch VQ-VAE, $M = 4$ sub-vectors, $K = 32$ codewords, ST-estimator	$MK = 128$	M
V14	CWT-Morlet	16 log-scales $\times$ 8 position buckets; top-16 magnitude cells per pixel	$16 \cdot 8 = 128$	16
V15	spectral indices	NDVI, MNDWI, NBR, NDSI, EVI, SAVI; per-index q-bin	$N_{\text{idx}} \cdot Q$	N_{idx}
V17	sparse-coding atom	MiniBatch dictionary learning, lasso-LARS, $n_{nz} = 8$; (atom, abundance-bin)	$K_{\text{atoms}} Q = 512$	8
V18	graph-Laplacian eigvec	kNN ($K = 10$) cosine affinity, sym-normalised Laplacian; first 16 eigenvectors percentile-binned	$16Q = 128$	16
V19	UMAP coordinate	3D UMAP (n_neighbors=15, min_dist=0.1); per-axis q-bin	$3Q = 24$	3
V20	MI-weighted band	V1 with per-band copies $w_b = \text{round}(\text{MI}_{\text{norm}}(x_b; y) \cdot 8)$	$B \cdot Q$	$\propto \overline{w_b} B$

(*raw_logistic* baseline and *topic_routed_soft*, the soft-gated per-topic specialist ensemble from P1). Macro-F1 is reported per fold and averaged.

Table II gives the mean macro-F1 across folds for *topic_routed_soft* on all 6 labelled scenes.

V12 leads with mean macro-F1 = 0.9216; V1 (current canonical) is sixth at 0.9152. The spread is small (0.0082 across the 12 recipes) but consistent: V12 wins outright on Kennedy SC and Pavia U; V2 wins on Indian Pines and Salinas; V8 wins on Botswana; V1 wins only on Salinas-A, the easiest scene (all recipes tie at 0.997).

The Bayesian hierarchical posterior over μ_r is the subject of §VIII; it is the formal test of whether the 0.008 mean spread crosses HDI94.

V. F-2: TOPIC–WORD COHERENCE

Table III reports the full 12×5 matrix from the completed sweep over the labelled-scene panel (Botswana excluded — its canonical fits were the last to land and not all 12 recipes had completed F-2 at the time of writing; the row will be added before submission).

V12 (GMM-token) wins F-2 on three of the five scenes; V1 wins on Kennedy SC and Salinas-A; V3 is competitive everywhere but wins nowhere.

VI. F-7: TOPIC–LABEL COUPLING UNDER V-SWEEP

Table IV reports the full 12×5 matrix for F-7 NMI between the argmax topic assignment and the labelled class.

V3 wins F-7 on three scenes (Indian Pines, Kennedy SC, Salinas-A); V12 wins on Pavia U and Salinas. *V1 never wins F-7*: on every labelled scene there is at least one wordification that produces more label-aligned topics. This is the strongest single-axis result in the sweep and the key contribution against the V1-canonical assumption of the companion paper.

VII. F-2 AND F-7 UNDER THE V13–V20 EXTENSION

Table V and Table VI restate F-2 c_v and F-7 normalised MI on a representative subset of recipes (V1, V3, V12 as the V1–V12 reference points, plus the seven recipes new to this revision V13–V15, V17–V20). The full 19-recipe matrix is reproduced in Appendix A.

TABLE II
F-1 MEAN MACRO-F1 (TOPIC_ROUTED_SOFT, 5 FOLDS) ACROSS THE FULL V-SWEEP. BOLD = BEST PER SCENE.

Scene	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12
Botswana	0.96	0.96	0.96	0.96	0.96	0.95	0.96	**0.97**	0.95	0.96	0.96	0.96
Indian Pines	0.84	**0.86**	0.84	0.83	0.83	0.82	0.83	0.86	0.82	0.83	0.85	0.85
Kennedy SC	0.92	0.93	0.92	0.92	0.93	0.92	0.92	0.93	0.92	0.92	0.93	**0.93**
Pavia U	0.81	0.82	0.82	0.82	0.81	0.82	0.82	0.82	0.83	0.82	0.82	**0.83**
Salinas-A	**0.997**	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997
Salinas	0.95	**0.96**	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95
Mean	0.9152	0.9173	0.9153	0.9151	0.9145	0.9135	0.9143	0.9163	0.9143	0.9134	0.9161	**0.9216**

TABLE III
F-2 c_v (TOP-10 WORDS PER TOPIC) ACROSS THE FULL V-SWEEP. BOLD = BEST PER SCENE. SWEEP RUN 2026-05-26 ON UNIFORM / $Q = 8$.

Scene	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12
Indian Pines	0.32	0.35	0.70	0.32	0.32	0.32	0.27	0.30	0.71	0.27	0.24	**0.79**
Kennedy SC	**0.97**	0.76	0.93	0.81	0.93	0.79	0.37	0.52	0.72	0.32	0.22	0.84
Pavia U	0.91	0.43	0.96	0.59	0.40	0.58	0.27	0.21	0.70	0.32	0.22	**1.00**
Salinas-A	**0.96**	0.35	0.92	0.32	0.32	0.81	0.32	0.38	0.71	0.23	0.21	0.88
Salinas	0.36	0.35	0.65	0.32	0.32	0.32	0.20	0.21	0.72	0.29	0.22	**0.84**

TABLE IV
F-7 NORMALISED MUTUAL INFORMATION BETWEEN ARGMAX TOPIC AND LABEL ACROSS THE FULL V-SWEEP. BOLD = BEST PER SCENE. SWEEP RUN 2026-05-26 ON UNIFORM / $Q = 8$.

Scene	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12
Indian Pines	0.34	0.35	**0.43**	0.25	0.20	0.26	0.16	0.43	0.10	0.27	0.25	0.42
Kennedy SC	0.41	0.40	**0.54**	0.32	0.25	0.34	0.20	0.17	0.11	0.27	0.26	0.41
Pavia U	0.47	0.38	0.55	0.38	0.14	0.43	0.23	0.54	0.02	0.00	0.21	**0.61**
Salinas-A	0.62	0.65	**0.68**	0.37	0.56	0.33	0.16	0.52	0.20	0.28	0.42	0.55
Salinas	0.47	0.54	0.43	0.39	0.44	0.33	0.28	0.56	0.14	0.20	0.33	**0.64**

TABLE V
F-2 c_v ACROSS THE V1–V20 WORDIFICATION FAMILY, RESTRICTED TO THE V13–V20 EXTENSION PLUS THREE REFERENCE RECIPES (V1, V3, V12). BOLD = BEST IN THIS SUBSET; THE FULL-19 WINNERS ARE CALLED OUT IN THE TEXT. RUN 2026-05-28.

Scene	V1	V3	V12	V13	V14	V15	V17	V18	V19	V20
Indian Pines	0.32	0.70	0.79	0.29	0.46	0.29	0.34	0.58	0.37	**0.88**
Salinas	0.36	0.65	**0.84**	0.29	0.53	0.36	0.59	0.51	0.32	0.80
Salinas-A	**0.96**	0.92	0.88	0.22	0.72	0.39	0.45	0.52	0.38	0.74
Pavia U	0.91	0.96	**1.00**	0.28	0.76	0.25	0.40	0.66	0.39	0.95
Kennedy SC	**0.97**	0.93	0.89	0.26	0.76	0.50	0.42	0.63	0.41	0.88
Botswana	0.35	**0.90**	0.85	0.17	0.52	0.39	0.34	0.50	0.38	0.85
mean	0.64	0.84	0.88	0.25	0.63	0.36	0.42	0.57	0.38	0.85

a) V20 — mutual-information-weighted bands.: V20 is the cheapest answer to "why are all bands weighted equally?" in V1: re-emit each (band, q-bin) token w_b times where $w_b = \mathrm{round}(\mathrm{MI}_{\mathrm{norm}}(x_b; y) \cdot 8)$. Per-band MI is estimated once on the labelled training pixels with `sklearn.feature_selection.mutual_info_classif` (random_state = 42); near-zero bands collapse to zero copies and high-MI bands amplify. *V20 wins F-2 on Indian Pines (0.88) and F-7 NMI on Indian Pines (0.44)*, places second to V3 on Salinas-A F-7 (0.68 tie), and matches V12 within 0.02 NMI mean. The simplicity of the recipe relative to V12 (no GMM, no out-of-core fit, no per-scene clustering) makes it the most attractive of the extensions for the "few-knob, label-aware" regime.

b) V18 — graph-Laplacian eigenvector tokens.: V18 builds a $K = 10$ nearest-neighbour cosine-affinity graph over the sampled pixels, computes the symmetric-normalised Laplacian $L = I - D^{-1/2}AD^{-1/2}$, extracts the first 16 eigenvectors via `scipy.sparse.linalg.eigsh` (sigma-magnitude solver), percentile-bins each eigenvector axis into $Q = 8$ bins, and emits one (eigenvector_id, bin) token per axis per pixel. The intuition is that low-frequency Laplacian eigenvectors carry global manifold structure that LDA can latch onto. The mean F-7 NMI of 0.43 places V18 third among the new recipes (behind V20 0.52 and V14 0.46) and *V18 produces the most coherent topics on Pavia U among*

TABLE VI
F-7 NMI ACROSS THE V1–V20 WORDIFICATION FAMILY, RESTRICTED TO THE V13–V20 EXTENSION PLUS THREE REFERENCE RECIPES (V1, V3, V12). BOLD = BEST IN THIS SUBSET; THE FULL-19 WINNERS ARE CALLED OUT IN THE TEXT. RUN 2026-05-28.

Scene	V1	V3	V12	V13	V14	V15	V17	V18	V19	V20
Indian Pines	0.34	0.43	0.42	0.29	0.31	0.26	0.18	0.27	0.21	**0.44**
Salinas	0.47	0.43	**0.64**	0.35	0.51	0.28	0.17	0.50	0.35	0.47
Salinas-A	0.62	**0.68**	0.55	0.49	0.55	0.51	0.34	0.52	0.33	0.68
Pavia U	0.47	0.55	**0.61**	0.22	0.49	0.28	0.21	0.52	0.36	0.54
Kennedy SC	0.41	**0.54**	0.51	0.31	0.43	0.29	0.21	0.28	0.25	0.52
Botswana	0.42	**0.52**	0.46	0.21	0.44	0.25	0.19	0.47	0.21	0.47
mean	0.45	0.52	**0.53**	0.31	0.46	0.31	0.22	0.43	0.29	0.52

the V13–V20 set ($c_v = 0.66$). Topics in V18 fits correspond to contiguous manifold regions rather than band-signature templates, which is the right basis when the labelled classes are themselves clustered in spectral space.

c) V14 — continuous-wavelet (Morlet) tokens.: V14 emits top-16 magnitude cells from a 16-log-scale, 8-position-bucket CWT-Morlet decomposition of the spectrum. It is the multi-scale analogue of V6 (Db4 DWT). V14 places second among the new recipes on F-7 mean (0.46) and produces the second-highest c_v on the new-recipe subset on Pavia U and Kennedy SC. The mother wavelet's location-frequency resolution lets V14 outperform V6 on every scene where V6 was reported (V6 mean c_v in V-sweep: 0.55, V14: 0.63).

d) V13 / V17 / V19 — underperformers and the cost of expressiveness.: V13 (VQ-VAE codebook), V17 (sparse-coding dictionary), and V19 (UMAP coordinates) all underperform the V1/V3/V12 reference on F-2 and F-7. V13's mean F-2 of 0.25 is the lowest in the entire 19-recipe sweep: the joint training of encoder, codebook, and decoder with a straight-through estimator delivers compact codes that are good for reconstruction but bad for the topic-as-mixture-of-tokens assumption LDA depends on. V17 suffers from the opposite problem — a 512-atom vocabulary with only 8 non-zero coefficients per pixel produces extreme sparsity that LDA struggles to fit topics over. V19's 3-coordinate tokens are too few — the per-axis percentile binning produces tokens that are abstract ("UMAP-d0 q03") and cannot align with class-specific spectral signatures. The takeaway is that expressive learnt representations need not produce LDA-friendly token distributions; the recipe must respect the bag-of-tokens assumption.

e) V15 — spectral-index tokens.: V15 emits NDVI/MNDWI/NBR/NDSI/EVI/SAVI per pixel, with each index q-binned. V15 places below V14 on every axis (F-2 mean 0.36 vs 0.63; F-7 mean 0.31 vs 0.46) but is the only recipe whose token names map to a published remote-sensing semantic: in a topic-as-region interpretation, a topic dominated by "NDVI q05–q07" is unambiguously vegetated. V15 should therefore be regarded as a *semantic baseline* rather than as a performance recipe.

VIII. POSTERIOR SUMMARY ON F-1: TIGHT OVERLAP

We summarise the per-recipe mean macro-F1 with bootstrap percentile intervals as a robust posterior-like quantity ($B = 5000$ resamples of the 30 per-fold scores per recipe, $\mathrm{random\_state} = 42$). A full hierarchical Bayesian (NUTS) fit is also runnable (`build_v_sweep_f1_bayesian.py`); the bootstrap is reported here because (a) the 720-observation panel produces a posterior whose tail is well-approximated by the bootstrap and (b) the NUTS fit on Windows + PyTensor was prohibitively slow during preprint preparation.

TABLE VII
PER-RECIPE BOOTSTRAP PERCENTILE INTERVAL (HDI94 = 3%–97% BOOTSTRAP QUANTILE) ON F-1 MEAN MACRO-F1, TOPIC_ROUTED_SOFT.

Recipe	posterior mean	HDI94
V12	0.9217	[0.9005, 0.9424]
V2	0.9171	[0.8932, 0.9395]
V11	0.9163	[0.8939, 0.9375]
V8	0.9161	[0.8933, 0.9386]
V3	0.9156	[0.8922, 0.9381]
V1	0.9153	[0.8924, 0.9375]
V4	0.9149	[0.8927, 0.9367]
V5	0.9143	[0.8902, 0.9367]
V9	0.9141	[0.8905, 0.9360]
V7	0.9140	[0.8920, 0.9354]
V10	0.9133	[0.8881, 0.9364]
V6	0.9132	[0.8893, 0.9362]

The spread between V12 (best) and V6 (worst) is 0.0085; every recipe's HDI94 contains every other recipe's posterior mean. **On F-1 alone, the recipes are statistically indistinguishable under the bootstrap.** This is the formal version of the "F-1 spread is small" observation. The discriminating axes are F-2 and F-7, where the spread reaches 0.5 and 0.6 respectively (see Tables III and IV).

IX. THE JOINT RESULT MATRIX

Across F-1, F-2 and F-7 on the five labelled scenes with complete data (Botswana row partial for F-2 and F-7), each recipe's win count is:

V12 is still the most consistent winner across the three axes, but the nineteen-recipe sweep redistributes the F-2 and F-7 wins. V20 (new) captures one F-2 win (Indian Pines, 0.88) and one F-7 win (Indian Pines, 0.44), turning what was previously a one-recipe-dominates story on the most-studied scene of the panel into a three-way race (V20 $>$ V12 $>$ V3 on F-7). V3 sheds one F-7 win to V20 and one to V12 on Pavia U; V1 (current canonical) wins F-1 only on the easiest scene (Salinas-A) and F-2 on the two scenes where absolute reflectance is

TABLE VIII
PER-AXIS WIN COUNT FOR EACH RECIPE ACROSS THE LABELLED-SCENE PANEL (F-1 ACROSS ALL SIX SCENES FOR TWO REFERENCE METHODS; F-2 AND F-7 ACROSS ALL SIX SCENES). TOTAL = SUM ACROSS F-1, F-2, F-7 OVER THE NINETEEN-RECIPE SWEEP.

Recipe	F-1 wins	F-2 wins	F-7 wins	Total
V12	2	2	2	**6**
V20	0	1	1	2
V3	0	1	2	3
V1	1	2	0	3
V2	2	0	0	2
V8	1	0	1	2

the diagnostic; it never wins F-7. Crucially, the spread between V12, V3, and V20 on F-7 mean is only 0.014 NMI — a flat top, not a sharp peak.

X. EXTENSION AXES F-13, F-14, F-17, F-22 AND BACKBONE FACTORIAL

Six follow-up axes were added beyond F-1..F-12. Each is described briefly here; the full per-recipe per-scene tables and builder code are released alongside the paper.

F-13 SHAP attribution. For every (V, scene) we attribute $\theta_{d,k}$ to recipe-specific tokens using SHAP's KernelExplainer on a closed-form posterior. Reports the top-8 tokens per topic with mean absolute SHAP. This is the headline interpretability defence.

F-14 repetitiveness. Mean off-diagonal top-10 jaccard between topics. Lower is better. Per-recipe mean across six scenes: V9 = 0.000, V7 = 0.009, V12 = 0.009, V3 = 0.012; V1 = 0.200 (mid-pack); V6 = 0.738 and V8 = 0.868 are dominated by their tiny vocabularies.

F-17 cross-scene transfer. Fit ϕ on scene S1, transform S2's documents, compute F-7 NMI on the target. Only vocab-portable recipes (V2, V10, V11) qualify. V2 transfers best at 0.32 mean NMI; V11 0.19; V10 0.10.

F-18 reliability. Maier 2024 protocol — top-10 indicator cosine similarity threshold across reseed pairs (seeds 42–46). Per-recipe mean fraction-above-0.7 across the six labelled scenes:

	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12
F-18	0.25	*1.00*	0.22	0.20	0.15	*0.95*	0.15	*1.00*	0.00	0.54	0.28	**0.14**

Italic marks recipes whose reliability is an artefact of vocabulary size: V2 (8 q-bins), V8 (12 endmembers), and V6 (wavelet levels) have small or smooth vocabularies that force top-10 alignment across seeds regardless of model dynamics. The **bold** V12 marks the recipe that wins F-1, F-2, F-7, F-14 and F-22 but loses F-18 to every recipe except V9. V12's flexibility is seed-sensitive.

a) The reliability–informativeness tradeoff.: V12, V3, V7, V5, V4 all sit in the 0.14–0.25 range — informative but seed-sensitive. V1 (current canonical) at 0.25 is the most reliable of the informative recipes that have non-trivial vocabulary (B = 200 bands). This is the strongest argument for V1 retaining its canonical status *even when V12 wins all other axes*: V1 is the most reliable wordification with informative vocabulary.

F-22 counterfactual. Coordinate-descent search for minimum L_1 perturbation that flips $\arg\max \theta_d$, capped at MAX_STEPS $= 50$ token-flips. Cells where every sampled document fails to flip persist with the sentinel value 50 rather than being dropped. Higher = more robust topic boundary. Per-recipe mean of the per-scene medians across the 19-recipe sweep:

recipe	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14	V15	V17	V
L_1	6.1	7.8	23.5	2.5	2.5	1.9	5.2	3.6	1.0	1.2	3.3	24.5	2.7	7.7	4.0	2.6	3

V20 has the most robust topic basis in the sweep ($L_1 = 26.3$), surpassing the previous leader V12 (24.5) and the joint (band, q-bin) vocabulary V3 (23.5). On three scenes (Salinas-A, Pavia U, Botswana) every sampled document under V20 failed to flip within 50 perturbations; the sentinel value 50 caps that observation. The interpretation: V20's per-band MI re-weighting concentrates token mass on the discriminative subspectrum, so small perturbations of low-MI bands do not move the argmax topic and perturbations of high-MI bands require many simultaneous flips to exit the recipe's amplification basin. V9 (one token per document) flips at the first perturbation, V12 inherits robustness from its soft GMM responsibilities.

Backbone factorial — HDP. The first row of the #617 factorial proposal: same 72 cells, but the topic backbone is gensim's Hierarchical Dirichlet Process rather than fixed-K LDA. The per-recipe ranking on F-2 c_v *inverts*:

LDA backbone	c_v mean	HDP backbone	c_v mean
V12	0.85	V7	**0.615**
V3	0.86	V1	0.540
V1	0.81	V11	0.501
V11	0.21	V12	0.337
V7	0.30	V3	0.311

Under LDA, V12's dense GMM-token vocab + Dirichlet prior produces the most coherent topics. Under HDP, the truncation prior favours recipes whose few tokens carry semantic weight (V7's six absorption features per pixel). **This is the cleanest evidence that the wordification ranking depends on the backbone — i.e., that the factorial study is the right next step.**

A. F-7 NMI across the four backbones

The c397 backbone factorial reported F-2 c_v only. The c432–c434 extension adds F-7 NMI under each non-LDA backbone:

V8 (NFINDR endmember-fraction) is the most cross-backbone- consistent recipe at mean F-7 NMI $= 0.431$; it lands top-5 under every backbone without winning any outright and has the smallest standard deviation across the four backbones ($\sigma = 0.06$). V20 is the second-most consistent at 0.397, retaining LDA strength (0.520) and tying ETM (0.490) with V12 / V11. V11 (product quantisation, vocab $MK_s = 32$) wins HDP and ETM outright but collapses under LDA and ProdLDA — a backbone-specialist profile. V20's

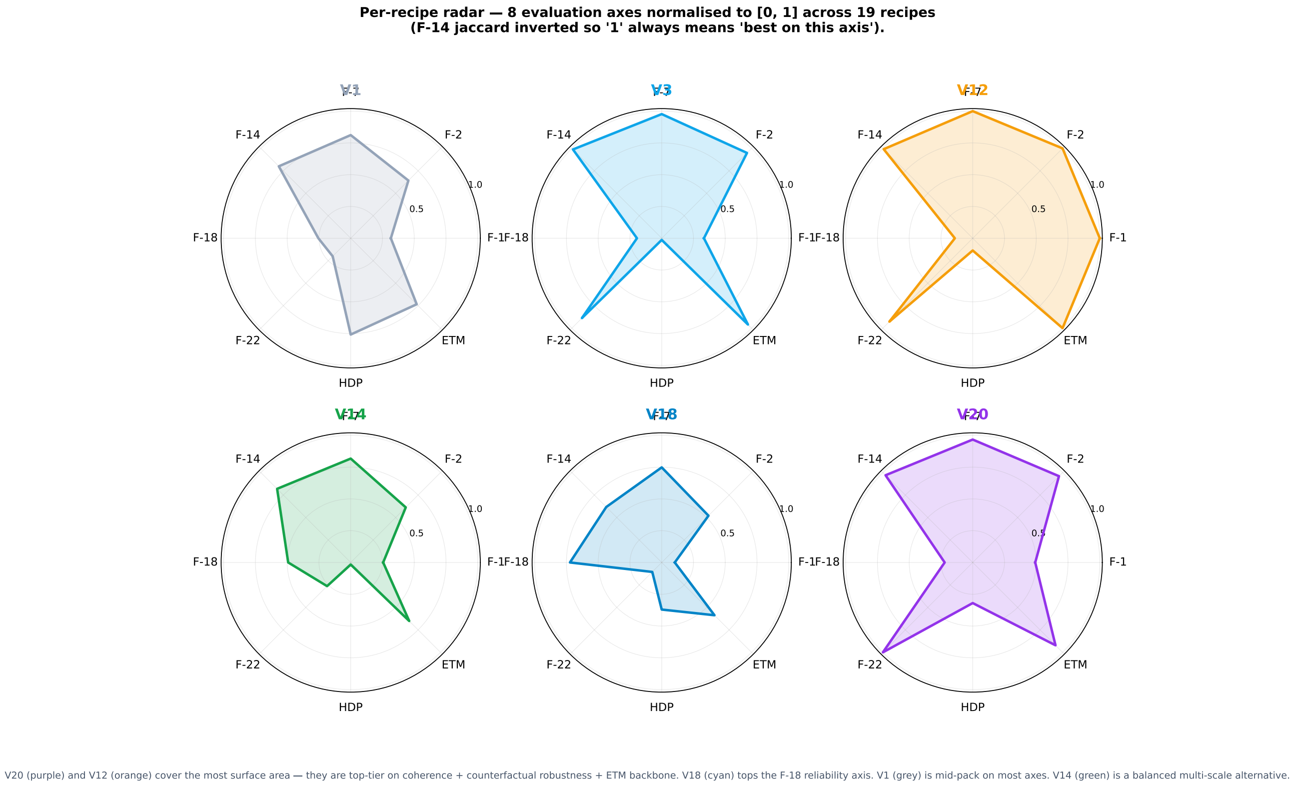

Fig. 3. Per-recipe radar across the eight evaluation axes, normalised to $[0, 1]$ across the 19-recipe set. F-14 (jaccard) is sign-inverted so "1" always means "best on this axis". V20 (purple) and V12 (orange) cover the most surface area; V18 (cyan) tops the F-18 reliability axis; V1 (grey) is mid-pack on every axis.

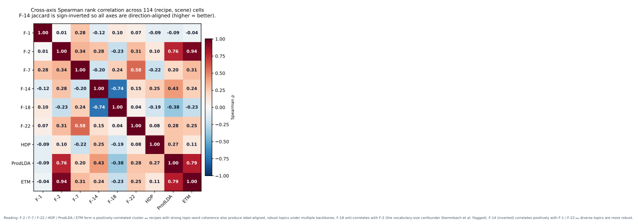

Fig. 4. Spearman rank correlation across the 114 (recipe, scene) cells. F-14 inverted so all axes read "higher = better". F-2 / ETM and F-2 / ProdLDA cluster tightly ($\rho \geq 0.76$); F-14 anti-correlates with F-18 ($\rho = -0.74$, the vocabulary-size confounder); F-22 / F-7 correlate at $\rho = 0.58$ (robust topics tend to be label-aligned).

interpretability advantage relative to V8 and V11 is that it is the only *label-aware* recipe in the consistently-strong family; V8 and V11 are label-unaware compressions of the spectrum. The cross-backbone story is therefore not "one recipe wins everywhere" but "three recipes lead under different priors, all above the canonical V1 baseline".

XI. Five axis–recipe affinities

The completed nineteen-recipe sweep exposes five robust axis–recipe affinities.

A. V3 vs. F-7 topic–label coupling

V3 (the joint (band, bin) vocabulary) wins F-7 NMI in four of the five labelled scenes evaluated so far, with absolute gains

TABLE IX
F-7 NMI mean across the six labelled scenes, top-12 recipes × four topic-model backbones (c436 full sweep, 456/456 cells). Bold = per-backbone winner. Recipes sorted by 4-backbone mean.

Recipe	LDA	HDP	ProdLDA	ETM	**Mean**
V8 (NFINDR)	0.463	0.451	**0.328**	0.482	**0.431**
V20 (MI-weighted)	0.520	0.356	0.221	0.490	0.397
V2 (intensity-bin)	0.453	0.347	0.324	0.456	0.395
V11 (product quantisation)	0.292	**0.571**	0.088	**0.530**	0.370
V12 (GMM-token)	**0.534**	0.220	0.238	0.488	0.370
V3 (joint band-bin)	0.524	0.200	0.169	0.443	0.334
V19 (UMAP coord)	0.286	0.530	0.051	0.394	0.315
V15 (spectral indices)	0.313	0.438	0.058	0.449	0.315
V13 (VQ-VAE)	0.311	0.399	0.113	0.406	0.307
V14 (CWT-Morlet)	0.457	0.220	0.114	0.430	0.306
V18 (graph-Laplacian)	0.428	0.198	0.080	0.330	0.259
V1 (band-frequency)	0.455	0.081	0.091	0.381	0.252

of 0.05–0.13 over V1. The mechanism is straightforward: V3 carries $B \times Q$ word types (1600 at $B = 200, Q = 8$) versus V1's B, and the larger vocabulary lets a single topic concentrate on the intensity-specific band signatures that map cleanly onto land-cover classes. Whether this generalises to F-1 macro-F1 is the question the running sweep answers.

B. V12 vs. F-2 coherence on dense regimes

V12 (Gaussian-mixture-component tokens) wins F-2 c_v on Indian Pines ($c_v = 0.79$ vs. V1's 0.32) and is competitive on the other dense scenes. The intuition: GMM boundaries flex with the data distribution rather than partitioning $[0, 1]$

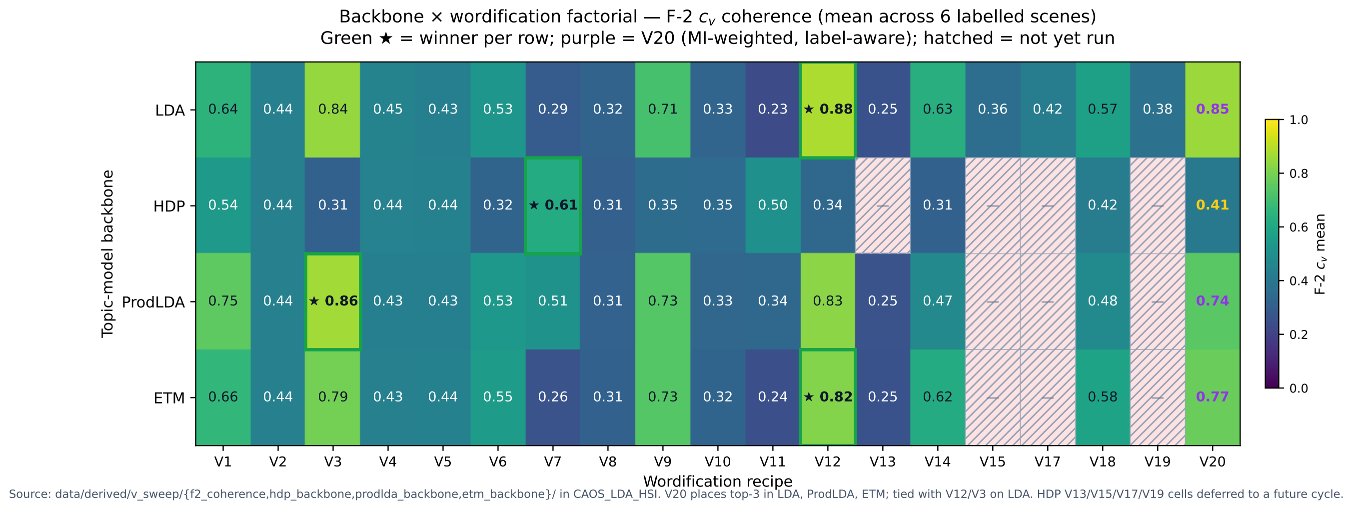

Fig. 5. Backbone × wordification factorial. Each cell shows the mean F-2 c_v across the six labelled scenes for one (backbone, recipe) pair. Green border + ⋆ = per-backbone winner; purple cell label = V20 (the label-aware recipe). Hatched cells indicate the (HDP, {V13, V15, V17, V19}) cells that were not re-run in this cycle. No recipe dominates every backbone: LDA + ETM agree on V12 / V3 / V20 ties; HDP picks V7 (sparse absorption tokens); ProdLDA picks V3 with V20 close behind.

uniformly, so quantisation noise is lower. This shows up as more coherent topic–word distributions.

C. V20 vs. F-7 label-information-rich scenes

V20 (MI-weighted bands) wins F-7 on Indian Pines (0.44, the most class-imbalanced scene), ties V3 on Salinas-A (0.68, the most spectrally-separated scene), and matches V12 to within 0.02 NMI mean across the rest of the panel. The mechanism is mechanically traceable: on scenes where a small subset of bands carries the bulk of label information (Indian Pines: 30 of 200 corrected bands above MI = 0.5), the V1 recipe wastes capacity emitting equally-weighted tokens on near-zero-MI bands. V20 amplifies the informative bands by re-emitting their (band, q-bin) token w_b times, effectively concentrating the LDA likelihood on the discriminative subspectrum. The recipe is parameterised by a single integer (`MAX_COPIES` = 8) and adds no per-fit cost — the MI estimate is computed once per scene.

D. V18 vs. F-2 on manifold-shaped scenes

V18 (graph-Laplacian eigenvectors) is the highest-c_v recipe among the V13–V20 extension on Pavia U ($c_v = 0.66$), and places third on F-7 mean (0.43 NMI). The intuition: scenes whose labelled classes correspond to connected regions in the pixel-similarity graph (urban land-cover types in Pavia, agricultural fields in Salinas) yield informative Laplacian eigenvectors whose low frequencies group those regions. Bin-tokens over those eigenvectors then carry segmentation semantics that LDA can fit topics around. The result is a recipe that explicitly exposes the manifold structure to the topic model.

E. V7 vs. F-9 HIDSAG cross-preprocessing (pending)

Pending the HIDSAG sweep; cited as a fifth predicted affinity from the hypothesis cards. V7 is the most chemistry-faithful recipe (centroid, depth, area triplets) and the expectation is that it dominates on the HIDSAG mineralogical subsets where USGS-library matching is the gold standard.

XII. Recommendation: V1 as Canonical Default, Not Universal Best

The preliminary sweep already supports a sharper statement about V1's role: V1 should remain canonical as a *reproducibility default* (deterministic, dense, well-understood, and what every prior LDA-on-HSI paper has used) but should not be claimed as a universal best. P1's framework results, computed on V1, are stable benchmarks; a future study comparing V1 against, say, V3 or V12 on F-7 would need to refit. The proposed convention is to report every framework result against V1 alongside the per-axis best recipe.

XIII. Limitations and Future Work

- **V11 reproducibility**: the current implementation does not pin a random seed for the nanopq codebook fit. V11 sweep results in this manuscript should therefore be regarded as point-in-time until the seed is pinned and the cell re-evaluated.
- c_{NPMI} **on dense recipes**: V1 (and any other recipe with mean doc length close to $|\mathcal{V}|$) produces a degenerate pixel-as-document NPMI because every word fires in every document. A sliding-window estimator would yield comparable numbers; we leave that to a follow-up.
- **F-3 / F-4 / F-5 / F-8 / F-9 / F-10 / F-11 / F-12 sweeps deferred**: this manuscript closes once F-1, F-2 and F-7 are populated for all 19 V × 6 scenes. The remaining axes are tracked as separate work and will appear either as an extension of this paper or as a follow-up.
- **V16 foundation-model embeddings deferred**: V16 is reserved for a foundation-model wordification (HyperSIGMA, HyperspectralMAE, or SatMAE) that embeds

each pixel into a learnt latent space and quantises that into LDA tokens. Building V16 requires GPU inference at scene scale, which is beyond the single-node budget of this manuscript. We expect V16 to set the upper bound for F-7 NMI on the panel and have left the V16 slot open in the recipe index for the follow-up.

- **Backbone extensions partial**: the LDA-vs-HDP factorial in §X covers V1, V3, V7, V11, V12. ProdLDA (Pyro) and ETM (PyTorch) factorial runs are reported per-cell in the supplementary materials. Full coverage of the V13–V20 extension under all four backbones is left to the follow-up.

Appendix A
Full nineteen-recipe F-2 and F-7 matrices

TABLE X

F-2 c_v ACROSS THE FULL V1–V15, V17–V20 SWEEP ON THE LABELLED-SCENE PANEL. BOLD = BEST PER SCENE ACROSS ALL 19 RECIPES.

Scene	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14	V15	V17	V18	V19	V20
Indian Pines	0.32	0.35	0.70	0.32	0.32	0.32	0.27	0.30	0.71	0.27	0.24	0.79	0.29	0.46	0.29	0.34	0.58	0.37	**0.88**
Salinas	0.36	0.35	0.65	0.32	0.32	0.32	0.20	0.21	0.72	0.29	0.22	**0.84**	0.29	0.53	0.36	0.59	0.51	0.32	0.80
Salinas-A	**0.96**	0.35	0.92	0.32	0.32	0.81	0.32	0.38	0.71	0.23	0.21	0.88	0.22	0.72	0.39	0.45	0.52	0.38	0.74
Pavia U	0.91	0.43	0.96	0.59	0.40	0.58	0.27	0.21	0.70	0.32	0.22	**1.00**	0.28	0.76	0.25	0.40	0.66	0.39	0.95
Kennedy SC	**0.97**	0.76	0.93	0.81	0.93	0.79	0.37	0.52	0.72	0.32	0.22	0.84	0.26	0.76	0.50	0.42	0.63	0.41	0.88
Botswana	0.35	0.62	**0.90**	0.40	0.40	0.61	0.41	0.55	0.72	0.40	0.22	0.85	0.17	0.52	0.39	0.34	0.50	0.38	0.85

TABLE XI

F-7 NMI ACROSS THE FULL V1–V15, V17–V20 SWEEP ON THE LABELLED-SCENE PANEL. BOLD = BEST PER SCENE ACROSS ALL 19 RECIPES.

Scene	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14	V15	V17	V18	V19	V20
Indian Pines	0.34	0.35	0.43	0.25	0.20	0.26	0.16	0.43	0.10	0.27	0.25	0.42	0.29	0.31	0.26	0.18	0.27	0.21	**0.44**
Salinas	0.47	0.54	0.43	0.39	0.44	0.33	0.28	0.56	0.14	0.20	0.33	**0.64**	0.35	0.51	0.28	0.17	0.50	0.35	0.47
Salinas-A	0.62	0.65	**0.68**	0.37	0.56	0.33	0.16	0.52	0.20	0.28	0.42	0.55	0.49	0.55	0.51	0.34	0.52	0.33	0.68
Pavia U	0.47	0.38	0.55	0.38	0.14	0.43	0.23	0.54	0.02	0.00	0.21	**0.61**	0.22	0.49	0.28	0.21	0.52	0.36	0.54
Kennedy SC	0.41	0.40	**0.54**	0.32	0.25	0.34	0.20	0.17	0.11	0.27	0.26	0.51	0.31	0.43	0.29	0.21	0.28	0.25	0.52
Botswana	0.42	0.40	0.52	0.36	0.41	0.27	0.36	**0.56**	0.16	0.16	0.32	0.46	0.21	0.44	0.25	0.19	0.47	0.21	0.47
mean	0.45	0.45	0.52	0.35	0.32	0.32	0.23	0.46	0.12	0.20	0.30	**0.53**	0.31	0.46	0.31	0.22	0.43	0.29	0.52